Human biology is one of the few domains where the amount of available data is beginning to exceed our ability to meaningfully integrate it. Medical records, lifestyle information, laboratory measurements, genomics, proteomics, and transcriptomics each provide a different view of the same underlying system, yet most models still treat them independently or only combine them in limited ways. Learning general-purpose representations from longitudinal, multimodal health data is an interesting open problem because it shifts the focus from solving individual prediction tasks toward building a broader understanding of biological state and change over time.

I have just completed my Master's in Data Science at Uppsala University, where I have gradually focused my interests on healthcare and machine learning. What initially drew me to the field was how data-driven systems can improve healthcare while remaining trustworthy and useful in practice. Early on, I wrote about bias in medical AI systems and how existing limitations in healthcare research risk being amplified as new, powerful models are deployed.

Through a health innovation course, I studied the European health technology ecosystem and the challenges involved in bringing new technologies into clinical use. In another project, I investigated healthcare data interoperability challenges in Sweden, examining how information moves between systems and organisations. Very recently, I started a pilot collaboration with the Haematology and Oncology Department at King Faisal Specialist Hospital in Saudi Arabia, exploring AI-assisted interpretation of electrophoresis results. While these experiences differ in scope and technical depth, they all reinforced the same observation: healthcare data is rich, fragmented, longitudinal, and captures different parts of the same biological processes.

My strongest technical experience comes from machine learning and software development. My master's thesis focuses on multimodal machine learning for harmful content detection, combining text, image, and contextual information within a unified modelling framework. The work required developing custom PyTorch architectures, designing representation learning approaches, handling large datasets, and making modelling decisions under significant ambiguity. Different domain, but key skills are transferable: integrating heterogeneous data sources, learning useful representations from incomplete information, and identifying which information is genuinely predictive versus merely correlated.

Most PhDs focus primarily on the research side of a problem. Here, the collaboration between ABC Labs and KTH creates an opportunity to follow the entire lifecycle of a project. From exploring research questions and developing new methods, to building systems around them and eventually seeing how useful outputs can become part of products used by real people.

The project itself also aligns well with how I think about healthcare data. Throughout my studies and projects, I have repeatedly encountered situations where valuable information exists across multiple disconnected sources. Clinical records, laboratory results, imaging, and other data streams each provide only a partial view of the underlying system. Access to large-scale datasets spanning medical records, lifestyle information, and multi-omics data makes this a rare opportunity to investigate those questions properly.

I’m currently on a job-seeking visa residing in Sweden, and unsure if that qualifies for this role.